

What was the Transformer initially introduced for?

The Transformer was initially introduced for **machine translation**.

The original architecture was designed to take one sequence as input and produce another sequence as output.

Example

English: I love machine learning

French: J'aime l'apprentissage automatique

So the initial use case was:

- **Machine translation**
- More generally, **sequence-to-sequence tasks**

Encoder-only Transformers

A well-known example is **BERT**.

Instead of primarily generating new text, encoder-only models are commonly used to **understand and represent existing text**.

This makes them useful for tasks such as:

1. Semantic Search
 1. Search based on **meaning**, rather than just matching exact words.
2. Document Retrieval
 1. Finding relevant documents from a large collection.
 2. Examples:
 - Company knowledge bases
 - Research papers
 - Legal documents
 - FAQs
 - Customer-support documents
3. Clustering
 1. Grouping similar pieces of text together

Decoder-only Transformers

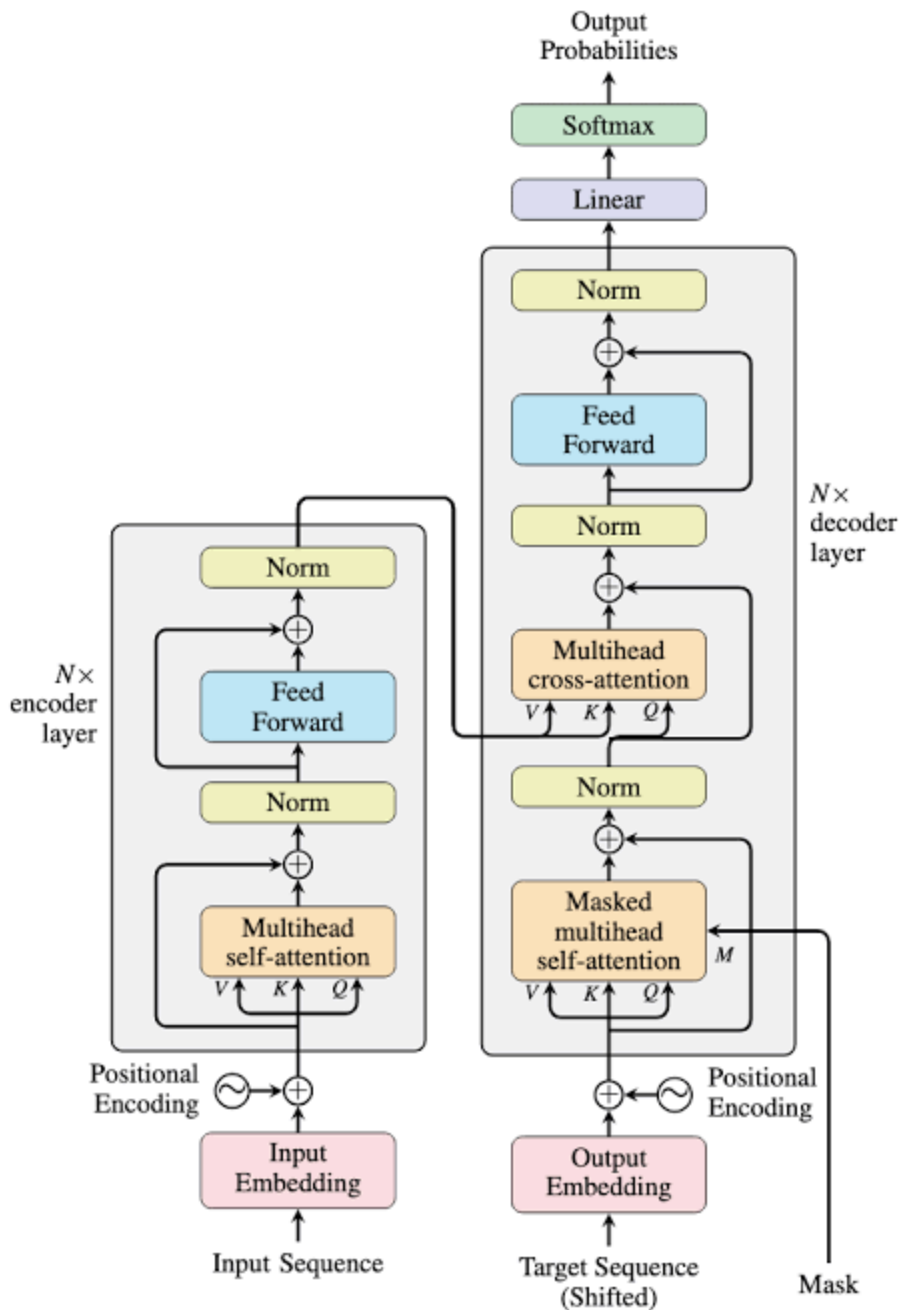
A major use of Transformers is **generation**.

Examples include GPT-style models.

They are commonly used for:

- Text generation
- Chatbots
- Code generation
- Question answering
- Content generation
- Summarization
- Writing assistance

Transformer Architecture



Important Concepts to learn about the transformers

1. Attention mechanism
2. Self-Attention
3. Cross-Attention
4. Multihead Attention
5. Positional Encoding
6. Mask Attention

Hands-On Lab session

We'll cover the following topic in the hands-on session:

- How to use transformers for machine translation
- How to use the encoder on model for sentiment analysis
- How to use the decoder only model for GPT type text generation

List of libraries and tools that are needed:

- Transformers library (and the models from huggingface)
- Google Colab (in case you don't want to run locally)

Running GPT type models Locally

We are going to run Ollama locally or on colab and see how to use them.

- Students should be asked to install ollama on systems or colab and give it prompts and check responses compared to GPT/ claude.

Instructions to install Ollama can be found in

- <https://ollama.com/download/linux>